Supplementary Information

Reimagining Psychiatric Care with Physical AI through Multiscale Modeling of Benefit Burden and Breakeven Conditions

Supplementary numerical contrasts and complete input provenance. The main manuscript contains all mathematical and operational Methods. Comprehensive parameter dictionaries are supplied as Supplementary Data 1–2, alongside reproducibility matrices and estimator diagnostics.

Supplementary display and data index

Supplementary Data index

Item	Description	Location
Supplementary Data 1	47 patient-model input, coefficient and structural rows; units, pathways, ranges, selection rationale, evidence class, citations and calibration status	supplementary_data_patient_parameters.csv
Supplementary Data 2	33 service-model input and structural rows with equivalent provenance fields	supplementary_data_service_parameters.csv
Supplementary Data 3	Full first/total-order Sobol results, bootstrap uncertainty and finite-design diagnostics	results/global_sensitivity/ sobol_first_total_indices.csv
Supplementary Data 4	All patient/service pairwise interaction indices and uncertainty	results/global_sensitivity/ sobol_second_order_indices.csv
Supplementary Data 5	All Saltelli input matrices and evaluated output matrices	results/global_sensitivity/raw/

Supplementary Table S1 | Alternative patient estimands

Table S1. Full endpoint estimand sensitivity

Outcome/estimand	Mean contrast	95% BCa interval	Replications
response locf percent	6.23	6.03 to 6.43	200
response bocf percent	5.53	5.33 to 5.73	200
response completers percent	10.21	9.96 to 10.47	200
remission locf percent	2.24	2.12 to 2.36	200
remission bocf percent	2.05	1.93 to 2.17	200
remission completers percent	3.54	3.37 to 3.71	200
mean change locf	-0.40	-0.41 to -0.38	200
mean change bocf	-0.25	-0.27 to -0.23	200
mean change completers	-0.84	-0.85 to -0.82	200

The endpoint estimand module uses a separate deterministic seed stream from the primary contrasts. Its independent paired-replication intervals describe variation under the same declared parameter settings.

Supplementary Table S2 | Service intake capacity

Table S2. Mean working-minute wait versus robot intake station count

Robot stations	Mean difference (minutes)	95% BCa interval
2	38.34	34.35 to 42.71
3	36.40	32.44 to 40.92
4	36.18	32.18 to 40.73
5	36.16	32.23 to 40.67

Screen intake capacity is fixed across this comparison. Independent deterministic seeds explain the small difference between the three-station sensitivity estimate and the primary three-station estimate.

Supplementary Table S3 | Pathway component diagnostics

Table S3. One-block-at-a-time model substitutions

Block	Outcome	Mean contrast	95% BCa interval
Embodiment symptom increment only	response difference percentage points	6.02	5.88 to 6.17
Embodiment symptom increment only	completion difference percentage points	0.01	-0.04 to 0.09
Embodiment symptom increment only	review minutes difference	-0.00	-0.01 to -0.00
Engagement dynamics only	response difference percentage points	2.59	2.36 to 2.80
Engagement dynamics only	completion difference percentage points	2.33	2.10 to 2.54
Engagement dynamics only	review minutes difference	0.01	0.00 to 0.02
Acceptance friction only	response difference percentage points	-1.85	-2.08 to -1.62
Acceptance friction only	completion difference percentage points	-7.13	-7.43 to -6.83
Acceptance friction only	review minutes difference	-0.10	-0.12 to -0.09
Reliability and fallback only	response difference percentage points	-0.32	-0.52 to -0.13
Reliability and fallback only	completion difference percentage points	-0.28	-0.50 to -0.07
Reliability and fallback only	review minutes difference	-0.01	-0.02 to 0.00
Safety oversight and physical exposure only	response difference percentage points	0.15	-0.13 to 0.42
Safety oversight and physical exposure only	completion difference percentage points	0.21	-0.07 to 0.51
Safety oversight and physical exposure only	review minutes difference	4.56	4.53 to 4.59
Full human-gated pathway bundle	response difference percentage points	6.38	6.11 to 6.65
Full human-gated pathway bundle	completion difference percentage points	-5.56	-5.86 to -5.25
Full human-gated pathway bundle	review minutes difference	4.36	4.33 to 4.38

These single-block substitutions are nonadditive pathway diagnostics. They use 120 repeated model trials with separately defined random streams and parameter substitutions.

Supplementary Table S4 | Repeated base-case configurations in distinct modules

Table S4. Independent Monte Carlo stream comparison

Metric	Primary experiment	Sensitivity experiment	Why values differ
Response (pp)	6.29	6.23 (alternative estimand); 6.38 (full bundle)	Different label-derived cohort and event seeds
Completion (pp)	-5.42	−5.40 (alternative estimand); −5.56 (full bundle)	Different label-derived cohort and event seeds
Review minutes/person	4.38	+4.36 (full bundle)	Independent bundle experiment
Service wait (min)	34.41	36.40 (3 stations)	Different service demand and latent event streams
Service throughput/day	-0.160	−0.178 (3 stations)	Different service demand and latent event streams

Primary and sensitivity estimates use separate Monte Carlo draws at identical specified parameter settings. Differences of this scale are consistent with ordinary finite-replication sampling variation.

Supplementary Table S5 | Model and deployment validation status

Table S5. Evidence needed before real-world decision use

Validation element	Current evidence	Prospective validation activity
Deterministic computational checks	Completed for the released model	Repeat automated checks following code changes
Fixed-assumption stochastic verification	Completed	Repeat stochastic verification following model changes
Structured stakeholder elicitation	Not completed	Recruit patients, clinicians, operations, robotics, privacy/human factors and modeling experts
Empirical parameter calibration	Not completed	Collect site, device and patient observational or trial data
Independent conceptual face validation	Not completed	Predefine acceptance criteria and record panel feedback
External predictive validation	Not completed	Evaluate model outputs against held-out clinical sites
Decision threshold specification	Not completed	Prespecify noninferiority margins and acceptable implementation risk

Supplementary Table S6 | Global estimator diagnostics

Table S6. Finite-design variance decomposition checks

Model	Output	Sum first-order estimates	Rows where estimated S1 > ST
individual	completion difference pp	0.858	3
individual	mean phq9 change difference	0.899	2
individual	missed crisis difference	0.204	0
individual	near miss difference	0.832	0
individual	response difference pp	0.737	0
individual	review minutes difference	0.932	1
service	abandonment difference pp	0.501	1
service	clinician utilization difference pp	0.964	2
service	false alert difference per 100	0.964	0
service	mean wait difference minutes	0.501	0
service	missed high acuity difference pp	0.965	0
service	p90 wait difference minutes	0.543	0
service	throughput difference per day	0.505	1

Exact Sobol contributions satisfy $0 \le S1 \le ST \le 1$ and $\Sigma S1 \le 1$. The table preserves finite-design estimates and provides their accompanying diagnostics; small inconsistent terms are excluded from substantive interpretation.

Source-specific parameter provenance

Supplementary Data 1–2 document each parameter by unit, functional role, reference pathway, range, evidence type, and calibration status. Literature citations are assigned where they support a specific scale, mechanism, or parameter-selection rationale. Numerical coefficients without an empirical source are explicitly designated as scenario assumptions. Display values are rounded consistently while executable configurations preserve full numerical precision.

Numbered citations in the parameter dictionaries correspond to the complete [1–48] reference list in the main manuscript. Preprints are identified as such in their bibliography entries.